

COMPUTER ORIENTED STATISTICS

Unit-III: Mathematical Expectation

Comprehensive Study Notes

3.1 Introduction to Mathematical Expectation

Theory

Mathematical Expectation (also called **Expected Value**) is a fundamental concept in probability theory. It represents the **average value** of a random variable over a large number of trials.

Key Definition

Mathematical Expectation of a random variable X is the weighted average of all possible values of X , where the weights are the probabilities.

For Discrete Random Variable

$$E(X) = \sum x_i \cdot P(X = x_i)$$

Where: x_i = Possible values of X , $P(X = x_i)$ = Probability of each value

For Continuous Random Variable

$$E(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

Where $f(x)$ is the probability density function.

3.2 Expected Value of a Function of a Random Variable

Theory

Often we need to find the expected value of a **function** of a random variable, not just the variable itself.

For Discrete Random Variable

If $Y = g(X)$ is a function of random variable X , then:

$$E[g(X)] = \sum g(x_i) \cdot P(X = x_i)$$

For Continuous Random Variable

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) \cdot f(x) dx$$

Common Functions and Their Expectations

Function	Expected Value	Name
$g(X) = X$	$E(X)$	Mean
$g(X) = X^2$	$E(X^2)$	Second Moment
$g(X) = (X - \mu)^2$	$\text{Var}(X)$	Variance
$g(X) = e^{tX}$	$M_X(t)$	Moment Generating Function
$g(X) = X^k$	$E(X^k)$	k-th Moment

Example 1: Function of Random Variable

Problem: Let X be the number of heads when tossing a fair coin twice. Find $E(X^2)$.

Solution: Sample space: {HH, HT, TH, TT}

X (Heads)	P(X)	X^2	$X^2 \times P(X)$
0	1/4	0	0
1	1/2	1	1/2
2	1/4	4	1

$$E(X^2) = \sum x^2 P(X) = 0 + 1/2 + 1 = 3/2$$

Example 2: Linear Function

Problem: If $E(X) = 5$, find $E(2X + 3)$.

$$E(2X + 3) = 2E(X) + 3 = 2(5) + 3 = 13$$

3.3 Properties of Expectation

Theory

Expectation has several important properties that make it a powerful tool in probability and statistics.

Property 1: Expectation of a Constant

If c is a constant, then:

$$E(c) = c$$

Example: $E(5) = 5$

Property 2: Expectation of cX

If c is a constant and X is a random variable:

$$E(cX) = c \cdot E(X)$$

Example: $E(3X) = 3E(X)$

Property 3: Expectation of $X + c$

$$E(X + c) = E(X) + c$$

Example: $E(X + 5) = E(X) + 5$

Property 4: Linearity (Additivity)

$$E(X + Y) = E(X) + E(Y)$$

Property 5: General Linearity

$$E(aX + bY + c) = aE(X) + bE(Y) + c$$

Property 6: Product of Independent Variables

If X and Y are independent:

$$E(XY) = E(X) \cdot E(Y)$$

Property 7: Monotonicity

If $X \leq Y$ for all outcomes, then:

$$E(X) \leq E(Y)$$

Property 8: Triangle Inequality

$$|E(X)| \leq E(|X|)$$

3.4 Examples on Properties of Expectation

Example 1: Using Linearity Property

Problem: Given $E(X) = 4$, $E(Y) = 6$, find $E(2X + 3Y - 5)$.

$$\begin{aligned} E(2X + 3Y - 5) &= 2E(X) + 3E(Y) - 5 \\ &= 2(4) + 3(6) - 5 \\ &= 8 + 18 - 5 = \mathbf{21} \end{aligned}$$

Example 2: Product of Independent Variables

Problem: X and Y are independent with $E(X) = 3$ and $E(Y) = 4$. Find $E(2XY - X + 3Y)$.

$$\begin{aligned} E(2XY - X + 3Y) &= 2E(XY) - E(X) + 3E(Y) \\ &= 2E(X)E(Y) - E(X) + 3E(Y) \\ &= 2(3)(4) - 3 + 3(4) \\ &= 24 - 3 + 12 = \mathbf{33} \end{aligned}$$

Example 3: Application of Properties

Problem: If $E(X) = 2$ and $E(Y) = 3$, find: (i) $E(3X + 2Y)$ (ii) $E(X - Y + 4)$

$$(i) E(3X + 2Y) = 3E(X) + 2E(Y) = 3(2) + 2(3) = 6 + 6 = \mathbf{12}$$

$$(ii) E(X - Y + 4) = E(X) - E(Y) + 4 = 2 - 3 + 4 = \mathbf{3}$$

3.5 Properties of Variance

Theory

Variance measures the spread or dispersion of a random variable around its mean.

$$\text{Var}(X) = \sigma^2 = E[(X - \mu)^2]$$

Alternative Formula

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Why Use This Alternative?

1. Easier to compute
2. Avoids negative values
3. Useful for theoretical derivations

Properties of Variance

Property 1: Variance of a Constant

If c is a constant:

$$\text{Var}(c) = 0$$

Property 2: Scaling Property

$$\text{Var}(cX) = c^2 \cdot \text{Var}(X)$$

Property 3: Translation Invariance

$$\text{Var}(X + c) = \text{Var}(X)$$

Property 4: Linear Combination (Independent Variables)

If X and Y are independent:

$$\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y)$$

Property 5: Standard Deviation

$$\text{SD}(X) = \sqrt{\text{Var}(X)}$$

Property 6: Chebyshev's Inequality

For any $k > 0$:

$$P(|X - \mu| \geq k\sigma) \leq 1/k^2$$

3.6 Complete Examples on Variance

Example 1: Finding Variance

Problem: Find the variance of the random variable X with distribution:

X	0	1	2	3
P(X)	1/8	3/8	3/8	1/8

Step 1: Find $E(X)$

$$E(X) = 0(1/8) + 1(3/8) + 2(3/8) + 3(1/8)$$

$$E(X) = 0 + 3/8 + 6/8 + 3/8 = 12/8 = 3/2 = 1.5$$

Step 2: Find $E(X^2)$

$$E(X^2) = 0^2(1/8) + 1^2(3/8) + 2^2(3/8) + 3^2(1/8)$$

$$E(X^2) = 0 + 3/8 + 12/8 + 9/8 = 24/8 = 3$$

Step 3: Find Variance

$$\begin{aligned} \text{Var}(X) &= E(X^2) - [E(X)]^2 = 3 - (3/2)^2 \\ &= 3 - 9/4 = 12/4 - 9/4 = \mathbf{3/4} \end{aligned}$$

Example 2: Using Properties of Variance

Problem: Given $\text{Var}(X) = 4$ and $\text{Var}(Y) = 9$, with X and Y independent, find: (i) $\text{Var}(2X)$ (ii) $\text{Var}(X + 5)$ (iii) $\text{Var}(2X + 3Y)$

$$(i) \text{Var}(2X) = 2^2 \times \text{Var}(X) = 4 \times 4 = \mathbf{16}$$

$$(ii) \text{Var}(X + 5) = \text{Var}(X) = \mathbf{4}$$

$$(iii) \text{Var}(2X + 3Y) = 4\text{Var}(X) + 9\text{Var}(Y) \\ = 4(4) + 9(9) = 16 + 81 = \mathbf{97}$$

Example 3: Complete Distribution Analysis

Problem: Find mean, variance, and standard deviation for:

X	1	2	3	4	5
P(X)	0.1	0.2	0.4	0.2	0.1

Step 1: Find E(X)

X	P(X)	X × P(X)
1	0.1	0.1
2	0.2	0.4
3	0.4	1.2
4	0.2	0.8
5	0.1	0.5
Total		3.0

$$E(X) = 3.0$$

Step 2: Find $E(X^2)$

X	X^2	$P(X)$	$X^2 \times P(X)$
1	1	0.1	0.1
2	4	0.2	0.8
3	9	0.4	3.6
4	16	0.2	3.2
5	25	0.1	2.5
Total			10.2

$$E(X^2) = 10.2$$

Step 3: Find Variance

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 10.2 - (3.0)^2 = 10.2 - 9 = \mathbf{1.2}$$

Step 4: Find Standard Deviation

$$\sigma = \sqrt{1.2} = \mathbf{1.095}$$

3.7 Covariance

Theory

Covariance measures the **joint variability** of two random variables. It indicates the direction of the linear relationship between them.

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Alternative Formula

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

Properties of Covariance

Property	Formula
Symmetric	$\text{Cov}(X, Y) = \text{Cov}(Y, X)$
Variance Special	$\text{Cov}(X, X) = \text{Var}(X)$
Constant	$\text{Cov}(c, X) = 0$
Scaling	$\text{Cov}(cX, dY) = cd \cdot \text{Cov}(X, Y)$
Additivity	$\text{Cov}(X+Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$
Linear	$\text{Cov}(aX+b, cY+d) = ac \cdot \text{Cov}(X, Y)$

Important Results

1. **If X and Y are independent:** $\text{Cov}(X, Y) = 0$
2. **If $\text{Cov}(X, Y) = 0$:** X and Y are uncorrelated (but NOT necessarily independent)

Variance of Sum

For any two random variables X and Y:

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

$$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) - 2\text{Cov}(X, Y)$$

3.8 Complete Examples on Covariance

Example 1: Finding Covariance

Problem: Find $\text{Cov}(X,Y)$ for the following joint distribution:

$X \setminus Y$	0	1
0	0.2	0.3
1	0.1	0.4

Step 1: Find marginal distributions

$$P(X=0) = 0.2 + 0.3 = 0.5$$

$$P(X=1) = 0.1 + 0.4 = 0.5$$

$$P(Y=0) = 0.2 + 0.1 = 0.3$$

$$P(Y=1) = 0.3 + 0.4 = 0.7$$

Step 2: Find $E(X)$

$$E(X) = 0(0.5) + 1(0.5) = 0.5$$

Step 3: Find $E(Y)$

$$E(Y) = 0(0.3) + 1(0.7) = 0.7$$

Step 4: Find $E(XY)$

X	Y	$P(X,Y)$	XY	$XY \times P$
0	0	0.2	0	0

0	1	0.3	0	0
1	0	0.1	0	0
1	1	0.4	1	0.4

$$E(XY) = 0.4$$

Step 5: Find Cov(X,Y)

$$\text{Cov}(X,Y) = E(XY) - E(X)E(Y) = 0.4 - (0.5)(0.7) = 0.4 - 0.35 = \mathbf{0.05}$$

Example 2: Using Covariance Properties

Problem: Given $\text{Var}(X) = 4$, $\text{Var}(Y) = 9$, $\text{Cov}(X,Y) = 2$, find: (i) $\text{Var}(X + Y)$ (ii) $\text{Var}(X - Y)$

$$\begin{aligned} \text{(i) } \text{Var}(X + Y) &= \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X,Y) \\ &= 4 + 9 + 2(2) = 13 + 4 = \mathbf{17} \end{aligned}$$

$$\begin{aligned} \text{(ii) } \text{Var}(X - Y) &= \text{Var}(X) + \text{Var}(Y) - 2\text{Cov}(X,Y) \\ &= 4 + 9 - 2(2) = 13 - 4 = \mathbf{9} \end{aligned}$$

Example 3: Covariance for Linear Functions

Problem: If $\text{Cov}(X,Y) = 3$, find $\text{Cov}(2X, 3Y)$.

$$\text{Cov}(2X, 3Y) = 2 \times 3 \times \text{Cov}(X,Y) = 6 \times 3 = \mathbf{18}$$

3.9 Relationship Between Measures

Summary Table

Measure	Formula	What It Measures
Mean	$E(X)$	Center of distribution
Variance	$E[(X - \mu)^2]$	Spread of distribution
SD	$\sqrt{\text{Var}(X)}$	Spread in original units
Covariance	$E[(X - \mu_X)(Y - \mu_Y)]$	Direction of relationship
Correlation	$\text{Cov} / (\sigma_X \sigma_Y)$	Strength of relationship

3.10 Complete Practice Problems

Problem 1: Complete Expectation Analysis

Problem: A random variable X has the following distribution:

X	-2	-1	0	1	2
$P(X)$	0.1	0.2	0.4	0.2	0.1

Find: $E(X)$, $E(X^2)$, $\text{Var}(X)$, and $\text{SD}(X)$.

Step 1: $E(X)$

X	P	$X \times P$
-2	0.1	-0.2
-1	0.2	-0.2
0	0.4	0
1	0.2	0.2
2	0.1	0.2
Total		0

$$E(X) = 0$$

Step 2: $E(X^2)$

X	X^2	P	$X^2 \times P$
-2	4	0.1	0.4

-1	1	0.2	0.2
0	0	0.4	0
1	1	0.2	0.2
2	4	0.1	0.4
Total			1.2

$$E(X^2) = 1.2$$

Step 3: Var(X)

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 1.2 - 0 = \mathbf{1.2}$$

Step 4: SD(X)

$$\sigma = \sqrt{1.2} = \mathbf{1.095}$$

Problem 2: Covariance and Variance of Sum

Problem: Given $E(X) = 2$, $E(Y) = 3$, $E(X^2) = 8$, $E(Y^2) = 13$, $E(XY) = 7$. Find: (i) $\text{Var}(X)$ (ii) $\text{Var}(Y)$ (iii) $\text{Cov}(X, Y)$ (iv) $\text{Var}(X + Y)$

$$(i) \text{Var}(X) = E(X^2) - [E(X)]^2 = 8 - 4 = \mathbf{4}$$

$$(ii) \text{Var}(Y) = E(Y^2) - [E(Y)]^2 = 13 - 9 = \mathbf{4}$$

$$(iii) \text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 7 - (2)(3) = 7 - 6 = \mathbf{1}$$

$$\begin{aligned} \text{(iv) } \text{Var}(X + Y) &= \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y) \\ &= 4 + 4 + 2(1) = \mathbf{10} \end{aligned}$$

Problem 3: Independent Variables

Problem: X and Y are independent with $E(X) = 4$, $\text{Var}(X) = 3$, $E(Y) = 2$, $\text{Var}(Y) = 5$. Find: (i) $E(3X - 2Y)$ (ii) $\text{Var}(3X - 2Y)$

$$\text{(i) } E(3X - 2Y) = 3E(X) - 2E(Y) = 3(4) - 2(2) = 12 - 4 = \mathbf{8}$$

$$\begin{aligned} \text{(ii) } \text{Var}(3X - 2Y) &= 9\text{Var}(X) + 4\text{Var}(Y) \\ &= 9(3) + 4(5) = 27 + 20 = \mathbf{47} \end{aligned}$$

3.11 Key Formulas Summary

Expectation Formulas

Type	Formula
Discrete	$E(X) = \sum x \cdot P(X=x)$
Continuous	$E(X) = \int x \cdot f(x) dx$
Function	$E[g(X)] = \sum g(x) \cdot P(X=x)$
Linearity	$E(aX + bY + c) = aE(X) + bE(Y) + c$
Product (Independent)	$E(XY) = E(X) \cdot E(Y)$

Variance Formulas

Type	Formula
Definition	$\text{Var}(X) = E[(X - \mu)^2]$
Alternative	$\text{Var}(X) = E(X^2) - [E(X)]^2$
Scaling	$\text{Var}(cX) = c^2\text{Var}(X)$
Translation	$\text{Var}(X + c) = \text{Var}(X)$
Sum (Independent)	$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$
Sum (General)	$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$

Covariance Formulas

Type	Formula
------	---------

Definition	$\text{Cov}(X,Y) = E[(X - \mu_X)(Y - \mu_Y)]$
Alternative	$\text{Cov}(X,Y) = E(XY) - E(X)E(Y)$
Scaling	$\text{Cov}(aX, bY) = ab \cdot \text{Cov}(X,Y)$
Special	$\text{Cov}(X,X) = \text{Var}(X)$

Previous Year Questions

Short Questions (2 marks)

1. Define mathematical expectation.
2. Write the formula for variance of a random variable.
3. What is covariance?
4. State the properties of expectation.
5. Write the alternative formula for variance.
6. What is the relationship between covariance and correlation?

Medium Questions (5 marks)

1. Explain the properties of mathematical expectation.
2. Find $E(X)$, $E(X^2)$, and $\text{Var}(X)$ for:

X	1	2	3	4
P(X)	0.2	0.3	0.4	0.1

3. Define covariance and state its properties.

4. If $E(X) = 3$, $\text{Var}(X) = 4$, find $E(2X+1)$ and $\text{Var}(2X+1)$.

Long Questions (12 marks)

1. (a) Define mathematical expectation and prove its properties.

X	-1	0	1	2
P(X)	1/8	3/8	3/8	1/8

2. (a) Define covariance and state its properties.

3. (a) Find $E(XY)$ for the joint distribution:

X \ Y	0	1	2
0	0.1	0.2	0.1
1	0.2	0.3	0.1

(b) Find $\text{Cov}(X,Y)$ and check if X and Y are independent.

Quick Reference Card

Key Theorems

Theorem	Formula
Expectation	$E(X) = \sum x \cdot P(X=x)$
Variance	$\text{Var}(X) = E(X^2) - [E(X)]^2$
Covariance	$\text{Cov}(X,Y) = E(XY) - E(X)E(Y)$
$\text{Var}(X+Y)$	$\text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X,Y)$

Important Properties

Property	Formula
Constant	$E(c) = c, \text{Var}(c) = 0$
Scale	$E(cX) = cE(X), \text{Var}(cX) = c^2\text{Var}(X)$
Shift	$E(X+c) = E(X)+c, \text{Var}(X+c) = \text{Var}(X)$
Linearity	$E(aX+bY+c) = aE(X)+bE(Y)+c$
Independence	$E(XY) = E(X)E(Y)$